Passive Acoustic Emission Monitoring of Plant Stress Responses: System Characterisation and Experimental Validation

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Abstract. We present the design, characterisation, and experimental evaluation of a low-cost passive acoustic emission (AE) monitoring system for plant physiological stress detection. The original system uses three piezoelectric channels; a Phase-3 follow-up replaces CH3/CH4 with MEMS microphones. All channels are amplified by LM358 circuits and recorded at 500 kµs in deep-memory mode (300 000 samples, 0.6 s per frame) on a Rigol DS1104Z. Frequency characterisation via a Raspberry Pi Pico chirp sweep identifies a calibrated piezo band of 5 kHz to 35 kHz (multi-frame coherence $\hat{\gamma}^2 \geq 0.95$). Multi-frame MLE time-delay estimation confirms equidistant sensor placement ($|\hat{\tau}_{\mathrm{MLE}}| < 0.3$ µs); a +1.95 µs offset at the 9.17 kHz steel eigenmode is a resonance-phase artefact. A controlled watering experiment yields no detectable AE after drift correction (Mann–Whitney $p = 0.86$). In the MEMS follow-up, moving both microphones from air into soil raises median RMS from 80.0 mV to 302.8 mV, yet both channels remain nearly simultaneous (GCC-PHAT median 0.000 µs) and share a 32.59 kHz line with $\hat{\gamma}^2 = 0.997$. No first–second-half band comparison survives FDR correction (0/32). The combined evidence indicates common-mode interference and one CH3 saturation fault, not plant AE. A positive-control reference experiment confirms pipeline integrity (52.6 dB sensitivity at 290 Hz, Wilcoxon $p < 10^{-12}$).

1. Introduction

Acoustic emission (AE) in plants (transient elastic waves generated by rapid internal stress redistributions) has been linked to xylem cavitation, cell-wall microfracture, and turgor-driven growth (Milburn and Johnson, 1966; Tyree and Dixon, 1983; Zweifel and Haesler, 2001). Detecting these signals non-invasively would provide a real-time proxy for plant water stress, relevant for precision irrigation and phenotyping. Classical AE plant research employs resonant ultrasonic sensors above 100 kHz (Cochard et al., 1992; Nardini et al., 2015; De Roo et al., 2016). More recent work extends AE sensing to airborne and contactless detection (Khait et al., 2023; Bonisoli et al., 2025; Klaminder et al., 2025), positions it as an early stress indicator for precision agriculture (Oletić et al., 2020; Király et al., 2025; Saha and Rastogi, 2026), and demonstrates first AE-driven irrigation prototypes (Oletić et al., 2023; Simbeye et al., 2023). Low-cost alternatives using standard op-amps and off-the-shelf oscilloscopes are attractive for continuous monitoring but their sensitivity limits are rarely rigorously quantified.

This work addresses that gap with five contributions: (i) full frequency-domain characterisation of a three-channel LM358-based acquisition chain; (ii) multi-frame MLE time-delay estimation for sensor placement verification; (iii) a statistically rigorous pre–post watering experiment with explicit drift correction; (iv) identification of the dominant noise sources limiting plant AE detectability in this class of hardware; and (v) a Phase-3 air-versus-soil evaluation of two MEMS channels using the same spectral, coherence, transient-screening, TDE, and FDR decision gates.

2. Nomenclature

Abbreviations.

AE	Acoustic emission
AC / DC	Alternating / direct current (input coupling / offset)
ADC	Analog-to-digital converter
DFT	Discrete Fourier transform
FDR	False discovery rate (Benjamini–Hochberg)
GBW	Gain–bandwidth product (of the amplifier)
MEMS	Micro-electro-mechanical system (microphone)
MLE	Maximum-likelihood estimation
PSD	Power spectral density (Welch estimate)
PZT	Lead zirconate titanate (piezo material)
RMS	Root mean square
SNR	Signal-to-noise ratio
TDE	Time-delay estimation
TTL	Transistor–transistor logic (sync pulse)

Symbols.

$\hat{\gamma}^2$	Magnitude-squared coherence between two channels (0–1; 1 = identical waveform)
$\hat{\tau}_{\mathrm{MLE}}$	Maximum-likelihood time-delay estimate (narrowband phase; Eq. 2)
$\hat{\tau}_{\mathrm{XC}}$	Cross-correlation time-delay estimate (cross-check)
$\hat{\tau}_{\mathrm{ph}}$	Phase-difference time-delay estimate
E_b	Mean band energy in a 5 kHz bin b
V_3, V_4	Signal of rod channels 3 and 4 (DFT X_n; RMS in Tab. 1)
f_0	Narrowband analysis frequency
f_{-3}	−3 dB (high-pass) corner frequency
e_n	Input-referred amplifier voltage-noise density ($\mathrm{nV}/\sqrt{\mathrm{Hz}}$)
N	Number of frames
M	Number of DFT bins within a band
α	Significance level (here 0.05)
τ	Kendall rank correlation (trend); *not* the time delay
p	Significance probability (p-value)
ΔE	Band-energy change, post − pre watering

3. Materials and Methods

3.1 Analysis Methods Overview

The AE characterisation pipeline combines four method families, applied in sequence from raw frames to statistical decision. They are detailed in the subsections that follow and summarised here for orientation.

- **Pre-processing.** Per-frame DC removal, linear detrending, Hann windowing, and a clipping/plausibility check reject saturated captures before analysis.
- **Spectral characterisation.** Averaged FFT and Welch PSD (nperseg = 8192) per channel; band energies E_b in 5 kHz bins (§3.3); prominence-based peak detection with mains-harmonic rejection, extracting frequency, prominence, bandwidth, Q-factor, and SNR per peak.
- **AE detection and validation.** Baseline-versus-active peak comparison (only peaks emerging under excitation are retained), repeatability across repeated captures, cross-channel consistency, and impulse-response resonance screening to separate sensor eigenmodes from genuine emissions.
- **Inter-channel and temporal analysis.** Multi-frame coherence $\hat{\gamma}^2$ (Eq. 1) and time-delay estimation by narrowband phase MLE with a cross-correlation cross-check (Eq. 2).
- **Statistical testing.** Wilcoxon signed-rank (paired pre–post), Mann–Whitney U (drift-corrected), Kendall τ (trend), all with Benjamini–Hochberg FDR control (§3.5).

3.2 Hardware Setup

Figure 1 shows the measurement chain. Channel 1 (CH1) is a bare soil piezo disc (27 mm) inserted into the root zone, biased through 830 kΩ and buffered by an LM358 amplifier. Channels 3 and 4 (CH3, CH4) use identical discs clamped to the ends of a 60 mm steel rod via 820 kΩ resistors and separate LM358 circuits; they serve as structural references. Channel 2 is defective and excluded. All channels use Rigol RP2200 passive probes at the 10× position. Probe compensation was performed manually against the oscilloscope's 1 kHz calibration square wave; oscilloscope self-calibration was not possible because the defective CH2 prevents freeing all inputs simultaneously.

The setup was subsequently relocated to another place and plant: CH1 remains a piezo sensor connected through its existing LM amplifier, whereas the piezo sensors on CH3 and CH4 were replaced by MEMS microphones connected through the existing LM amplifier stages. The monitored specimen is a lettuce plant (*Lactuca sativa*) rather than the *Euphorbia leuconeura* used previously. Sections 3–5 retain the historical three-piezo results on *Euphorbia leuconeura*; Section 6 reports the separately characterised MEMS air-reference and soil-insertion runs on *Lactuca sativa*.

Table 1: Calibration band summary (Pico chirp sweep). Band in kHz; V_3, V_4 are per-band RMS amplitudes (mV). Reliable: $\hat{\gamma}^2 \geq 0.9$.

Band	$\hat{\gamma}^2$	V_3	V_4	$\hat{\tau}_{\mathrm{MLE}}$ (µs)
5	0.962	6.44	6.42	-0.26
10	1.000	105.7	98.6	+0.19
15	0.993	11.3	10.6	-0.20
20	0.976	32.6	31.0	-0.06
25	0.983	10.8	10.2	-0.67
30	0.996	53.4	51.7	-0.08
35	0.959	14.5	14.2	+0.02
40	0.064	4.15	3.92	+4.29
50	0.065	3.94	3.01	+2.11

3.3 Data Acquisition and Processing

Measuring at 500 kSa/s, 300 000 samples (0.6 s), AC coupling, 0.5 V/div. A Raspberry Pi Pico generates a 20 kHz to 100 kHz chirp (30 s sweep) for calibration, synchronised to the oscilloscope via a 500 µs TTL pulse at every 5 kHz milestone.

Each acquisition automatically commits its raw waveforms (`.npz`) to the project repository via git-LFS, alongside a per-run JSON/Markdown summary and metadata. This yields a versioned, append-only raw-data pipeline in which every reported figure is traceable to an immutable, content-addressed recording.

Band energies: $E_b = \frac{1}{NM}\sum_{n=1}^{N}\sum_{m\in b}|X_n(m)|^2$, where X_n is the DFT of frame n and b is a 5 kHz-wide bin. Multi-frame coherence:

$$\hat{\gamma}^2(f) = \frac{\left|\sum_n V_3(n,f)\, V_4^*(n,f)\right|^2}{\sum_n |V_3|^2 \cdot \sum_n |V_4|^2}. \tag{1}$$

3.4 Time-Delay Estimation

MLE at narrowband frequency f_0:

$$\hat{\tau}_{\mathrm{MLE}} = -\frac{\angle\left(\sum_n V_3(n,f_0)\, V_4^*(n,f_0)\right)}{2\pi f_0}, \tag{2}$$

unambiguous for $|\tau| < (2f_0)^{-1}$. A cross-correlation estimate $\hat{\tau}_{\mathrm{XC}}$ provides a cross-check.

3.5 Statistical Methods

Pre–post comparisons: Wilcoxon signed-rank test on paired frames. Multiple-band comparisons: Benjamini–Hochberg FDR at $\alpha = 0.05$ (Benjamini and Hochberg, 1995). Trend detection: Kendall τ. Drift-corrected test: Mann–Whitney U comparing pre-watering steady-state (frames 0–4) against post-watering steady-state (frames 10–19).

4. System Characterisation

4.1 Amplifier Bandwidth and Coherence

Figure 2 shows results from the Pico chirp sweep. Three regions are identified:

- 5 kHz **to** 35 kHz: $\hat{\gamma}^2 \geq 0.95$, flat amplitude except at resonance peaks (9.17 kHz, 20 kHz, 30 kHz steel eigenmodes). *Calibrated measurement band.*

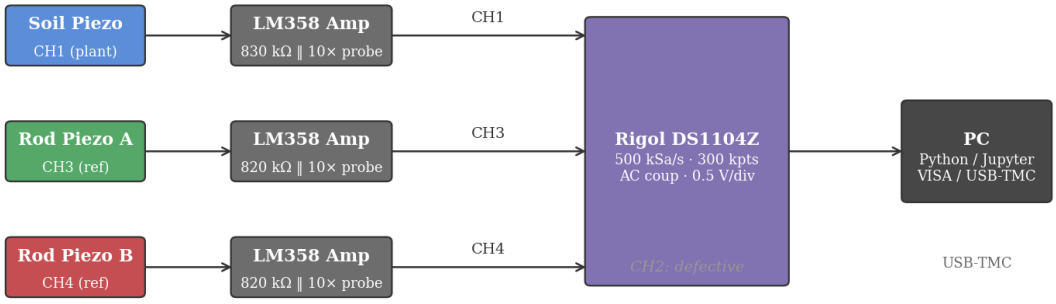


Figure 1: Main measurement system used to acquire the data analysed in this paper. Three piezo channels amplified by LM358 circuits are recorded by a Rigol DS1104Z at 500 kSa/s, 300 000 pts per frame.

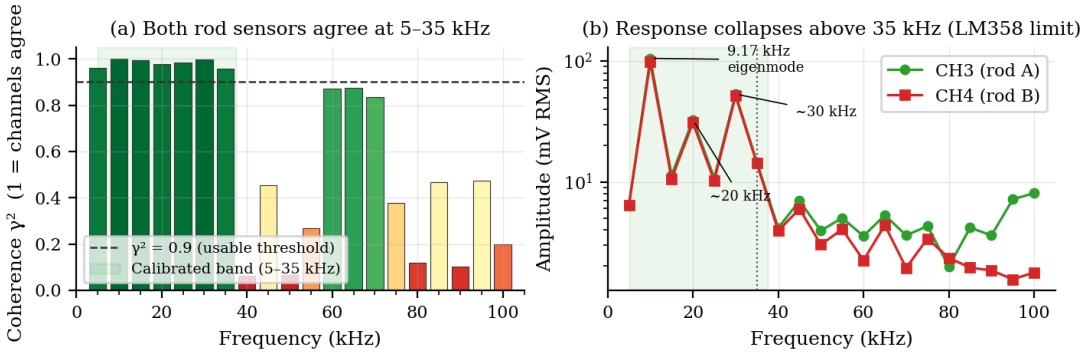


Figure 2: How faithfully the chain passes each frequency. A Raspberry Pi Pico injects a known 20 kHz to 100 kHz chirp; the two rod sensors (CH3, CH4) record it. (a) Coherence $\hat{\gamma}^2$ between the two channels per 5 kHz band (1 = identical waveform, i.e. the frequency is passed cleanly); green shading marks the calibrated band. (b) Recorded RMS amplitude of each channel; the peaks at 9.17 kHz, 20 kHz, 30 kHz are steel-rod eigenmodes.

- 40 kHz **to** 55 kHz: $\hat{\gamma}^2 = 0.06$–0.45; LM358 roll-off.
- 60 kHz **to** 75 kHz: partial recovery ($\hat{\gamma}^2 \approx 0.85$), tentatively spurious probe-cable resonance; unreliable.

The LM358 with 830 kΩ parallel input resistance and estimated piezo capacitance 10 nF yields high-pass corner $f_{-3} \approx 19$ Hz, leaving all mains harmonics ≥ 50 Hz unattenuated.

4.2 Time-Delay Estimation

Figure 3a shows $\hat{\tau}_{\mathrm{MLE}}$ for all 20 calibration bands. Within the calibrated band, all delays lie within ±0.3 µs, confirming equidistant sensor placement on the rod. Unreliable bands ($\hat{\gamma}^2 < 0.5$) show scatter up to ±4.3 µs, demonstrating that low coherence directly inflates TDE variance.

At the 9.17 kHz eigenmode, fine-resolution phase-difference estimation across $N = 15$ clean frames yields $\hat{\tau}_{\mathrm{ph}} = +1.950(31)$ µs and $\hat{\tau}_{\mathrm{XC}} = +1.60$ µs (Fig. 3b). Both are consistent and physically interpretable as a resonance-induced group-delay offset (Knapp and Carter, 1976): at the eigenfrequency the rod acts as a delay line for the propagating mode, introducing a geometry-independent phase difference between sensor positions.

4.3 Channel Noise Floor

Passive band-energy spectra from the pre-watering baseline ($N = 10$ frames) reveal three distinct channel noise regimes.

CH4 (rod piezo B) shows a flat spectrum of ≈ 0.8 mV RMS per 5 kHz band, totalling 25.7 mV RMS. This matches the LM358 noise prediction ($40\,\mathrm{nV}/\sqrt{\mathrm{Hz}} \times \sqrt{500\,\mathrm{kHz}} \approx 28$ mV), confirming CH4 as amplifier-thermal-noise limited: the fundamental hardware floor.

CH3 (rod piezo A) shows 39 dB excess in the 5 kHz to 10 kHz band (68 mV RMS), from ambient coupling into the 9.17 kHz steel eigenmode even without active excitation. CH3 is paradoxically *worse* than CH4 as a passive AE reference because structural resonance dominates.

CH1 (soil piezo) carries 137 mV total RMS, of which 129 mV are in the 0 kHz to 5 kHz band. Contributing mechanisms: (i) LM358 $1/f$ noise (dominant below 1 kHz); (ii) capacitive mains coupling through soil; (iii) ambient vibrations via soil matrix. AC coupling at the oscilloscope input ($f_{-3} < 1$ Hz) does not attenuate any of these.

5. Plant AE Experiment

5.1 Protocol

1. **Baseline** ($N_{\mathrm{pre}} = 10$ frames, 0.6 s each): continuous recording.
2. **Watering**: single 10 s drip event via microcontroller.
3. **Response** ($N_{\mathrm{post}} = 20$ frames): continuous recording immediately after.

Total window: 30 min. Plant: potted *Euphorbia leuconeura*, ≈ 18 months.

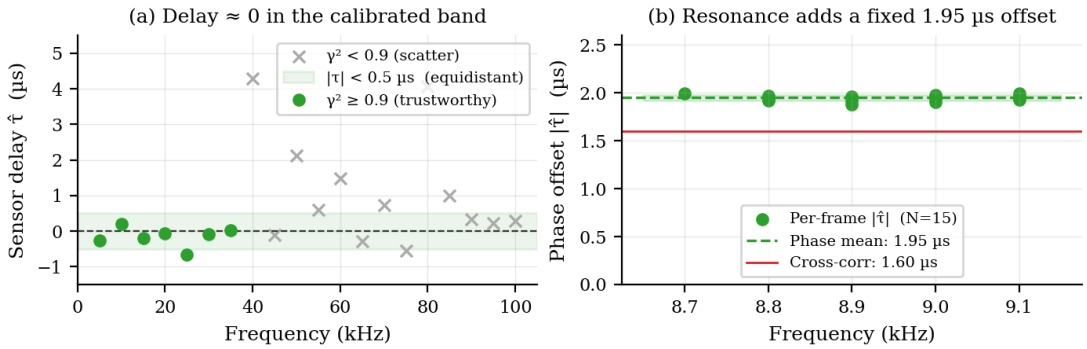


Figure 3: Time-delay between the two rod sensors. (a) MLE delay $\hat{\tau}_{\mathrm{MLE}}$ per 5 kHz band; green circles are trustworthy ($\hat{\gamma}^2 \geq 0.9$), grey crosses are low-coherence bands whose scatter is meaningless. In the calibrated band all delays sit inside ± 0.5 µs (green band), i.e. the sensors are equidistant. (b) Zoom on the 9.17 kHz steel resonance ($N = 15$ clean frames): every frame gives the same ≈ 1.95 µs offset, a fixed resonance-phase artefact rather than a geometric path difference. Magnitudes shown; the phase and cross-correlation estimates agree to within 0.35 µs.

5.2 Pre–Post Watering Analysis

A monotone upward trend is present within both the pre-watering sequence (Kendall $\tau = +0.69$, $p = 0.005$) and the post-watering sequence ($\tau = +0.90$, $p < 10^{-4}$) , consistent with thermal warming of the amplifier over time. The apparent -2.9 mV drop at the watering boundary is a mechanical transient from physical disturbance, not an acoustic emission event.

The uncorrected Wilcoxon test (paired, pre vs. post) yields $p = 0.002$ for CH1 RMS, nominally significant. This is a *temporal confound*: the test compares pre-watering frames at the end of the warm-up curve (138.5 mV) against post-watering frames at the start of a new warm-up cycle (135.6 mV). After drift correction, comparing pre-steady-state (frames 0–4) against post-steady-state (frames 10–19), Mann–Whitney yields $p = 0.86$, with means 137.36 mV and 137.42 mV respectively, a difference of 0.06 mV.

Band-level FDR analysis nominally flags eight bands, but their directions are inconsistent (3 positive, 5 negative) and effect sizes negligible ($\max|\Delta E| = 3.0$ dB). For comparison, the positive control in Section 7 shows 52.6 dB sensitivity. The two most significant bands (50 kHz to 55 kHz) fall in the amplifier roll-off zone ($\hat{\gamma}^2 = 0.06$) and are uninformative.

Result: No plant AE signal is detectable. The 137 mV noise floor on CH1 exceeds any credible plant AE amplitude by at least one order of magnitude.

5.3 Continuous Monitoring

A 51-hour continuous session logged 817 persistent cross-channel peak events. The dominant peak at 1.5 kHz (30th harmonic of 50 Hz mains) appears on all three channels simultaneously in 62% of events, i.e. a common environmental source. Only 40 events (4.9%) appear on CH1 but not CH3, the subset most likely to be soil-mediated; they cluster near 6 kHz with no biological timing pattern, consistent with the null watering result.

6. MEMS Characterisation

6.1 Frequency Calibration and Noise Floor

Before the passive campaigns, the two MEMS channels (CH3, CH4) were characterised directly from raw data with the same estimators as the piezo chain (Sec. 3.2). A Pico sweep (5 kHz to 100 kHz, 5 kHz steps, 80 valid frames) drives the coherence, amplitude and MLE time-delay analysis (Fig. 4a,b), recomputed from the raw complex FFT bins; a passive deep-memory run gives the noise floor (Fig. 4c).

The calibrated band ($\hat{\gamma}^2 \geq 0.9$ and SNR ≥ 3 on both channels) again spans 5 kHz to 35 kHz, bounded by the LM roll-off, with the strongest transfer at 10 kHz and 30 kHz. CH3 and CH4 track each other closely; across the reliable bands the sweep MLE time-delay stays within about 1 µs (excluding the low-SNR 5 kHz point), consistent with near-equidistant placement.

6.2 Protocol and Decision Gates

First an air reference was conducted just above the soil and then a follow-up with both microphones inserted into the soil at unequal distances from the presumed plant source region. Each run contains $N = 20$ synchronised frames at 500 kSa/s and 300 000 samples per channel.

The notebook method chain was applied without changing its decision logic: DC removal and linear detrending; clipping/plausibility rejection; Welch PSD (nperseg = 8192); 5 kHz band energies and prominence peaks; robust Hilbert-envelope transient screening in 20 kHz to 100 kHz; CH3–CH4 coherence; narrowband phase MLE with GCC-PHAT cross-check; Kendall trend tests; and first-half versus second-half Mann–Whitney tests with Benjamini–Hochberg FDR. A credible plant event had to be transient rather than periodic, exceed the passive baseline, appear plausibly on both sensors, and exhibit a non-zero delay consistent with the unequal geometry.

6.3 Air Reference and Soil Results

In air, median RMS was 80.67 mV on CH3 and 79.38 mV on CH4 (CH4/CH3 = -0.25 dB). Seventeen of 20 frames lay within ± 1 µs; after rejecting three low-

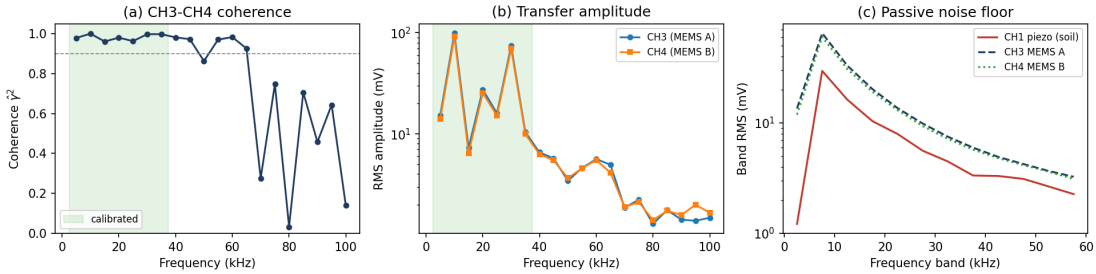


Figure 4: MEMS frequency calibration and noise floor. (a) CH3–CH4 coherence $\hat{\gamma}^2$ per band; green shading marks the calibrated 5 kHz to 35 kHz band. (b) Recorded RMS transfer amplitude of both MEMS channels (peaks at 10 and 30 kHz). (c) Passive noise floor: the two MEMS channels (CH3 81 mV, CH4 75 mV total AC RMS) sit above the soil piezo (CH1 39 mV) and are dominated by the common-mode comb in the 5 kHz to 10 kHz band.

confidence boundary solutions, the GCC-PHAT median was 0.000 µs. Phase candidates occurred mostly at the upper analysis boundary (90 kHz to 100 kHz) and had 196 µs to 322 µs uncertainty, far exceeding their fitted delays.

Soil insertion increased the median RMS to 306.79 mV (CH3) and 298.71 mV (CH4), while the relative level remained close (CH4/CH3 = −0.23 dB). The spectrum is a dense line comb shared by both channels; its strongest line at 32.593 kHz has coherence $\hat{\gamma}^2 = 0.997$. Mean 20 kHz to 100 kHz coherence is 0.603, and 14/16 five-kilohertz bands exceed 0.7. Nevertheless, all 20 GCC-PHAT estimates are exactly 0.000 µs; phase-MLE delays span only −0.017 µs to +0.065 µs. This near-zero delay is inconsistent with using the unequal sensor distances to localise a soil-borne source and instead indicates common electrical or environmental coupling.

Transient screening produces approximately 544 threshold crossings per channel in every 0.6 s frame, with 10,392 CH3 candidates matched on CH4 within 200 µs. Their periodicity rejects them as AE clicks. Frame 13 contains the only large channel-specific excursion: CH3 steps from approximately +1.4 V to −2.54 V and remains at the rail while CH4 stays normal. The discontinuity and sustained plateau identify saturation/overflow, not an elastic transient. No first–second-half band comparison survives FDR correction (0/32); RMS trends are non-significant on CH3 (Kendall $\tau = -0.253$, $p = 0.128$) and nominal on CH4 ($\tau = -0.358$, $p = 0.028$) but unsupported by any FDR-significant spectral change.

Result: both MEMS channels respond and have similar average sensitivity, but neither the air nor soil run provides evidence of plant AE. Soil insertion mainly amplifies a coherent periodic interference field.

7. Positive Control: Reference Channel Experiment

To verify pipeline integrity, a reference experiment ($N = 40$ frames) was conducted with the Pico sweep driving CH1 while CH2 captured the same signal and CH3 served as a fully passive reference (Fig. 5).

CH1–CH2 multi-frame coherence at 290 Hz reaches $\hat{\gamma}^2 = 0.928$, versus $\hat{\gamma}^2 \approx 0.01$ for CH1–CH3, a 52.6 dB sensitivity advantage. Wilcoxon signed-rank on per-frame RMS (CH1 vs. CH3) gives $p < 10^{-12}$. This confirms that the statistical pipeline detects genuine correlated signals when present, and that the negative watering result reflects a true absence of detectable AE above the noise floor.

Table 2: Noise floor and estimated SNR. Thermal limit: $e_n\sqrt{B}$, $e_n = 40\,\text{nV}/\sqrt{\text{Hz}}$. AE range from Cochard et al. (1992).

Ch.	RMS (mV)	Band	Noise source
CH4	25.7	Wideband	LM358 thermal
CH3	84.0	5–10 kHz	Steel eigenmode
CH1	137.0	0–5 kHz	$1/f$ + mains

Plant AE (cavitation): 0.01–1 mV RMS (Cochard et al., 1992); SNR on CH1: −43 to −23 dB (undetectable).

8. Discussion

8.1 Noise Floor as Primary Limitation

CH1's 137 mV floor (LM358 $1/f$ noise plus weak mechanical isolation) is ≈ 80 dB above a low-noise preamplifier ($e_n \approx 4\,\text{nV}/\sqrt{\text{Hz}}$, e.g. OPA2134). Reported plant AE amplitudes span tens of microvolts (single cavitation, Cochard et al., 1992) to a few millivolts, i.e. 20 dB to 40 dB below this floor. The LM358 also rolls off at 35 kHz, excluding the primary cavitation band (100 kHz to 500 kHz); the usable 5 kHz to 35 kHz window, where slow hydraulic AE is reported (Gagliano et al., 2012; Khait et al., 2023), is noisy.

8.2 Structural and Thermal Artefacts

The passive 39 dB excess on CH3 at 9.17 kHz shows how a structural resonance amplifies ambient vibration straight into the band of interest; a damped reference mount or anechoic enclosure would remove it. Separately, the reproducible +1.5 mV warm-up drift confounds sequential pre–post tests: future protocols should stabilise for 30 min, interleave pre/post frames, and always report Kendall τ next to any Wilcoxon result on sequential data.

The MEMS follow-up reaches the same null decision by a different failure mode. Soil insertion raises both channels by approximately 11.6 dB relative to the air reference, but the almost identical spectra, 32.59 kHz coherence of 0.997, and zero inter-channel delay show that the added energy is not a spatially propagating plant signal.

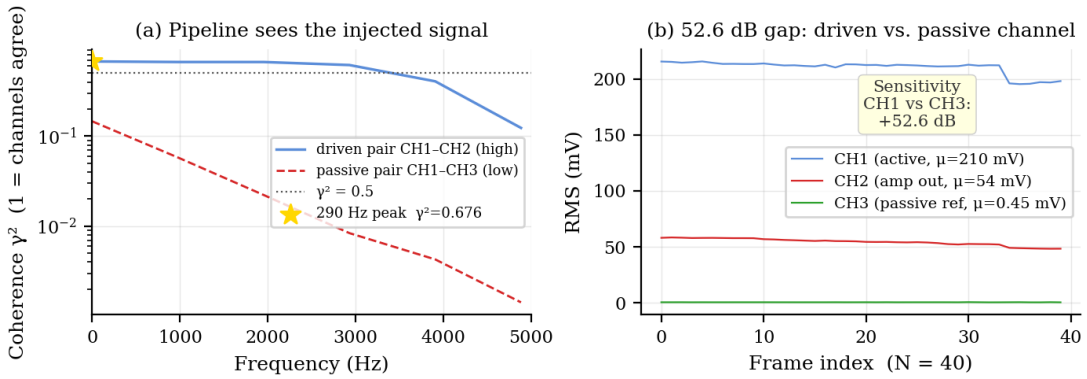


Figure 5: Positive control: a known Pico signal is fed into CH1 ($N = 40$ frames). (a) Coherence between the driven pair (CH1–CH2) and the passive pair (CH1–CH3); the driven pair tracks the signal (peak $\hat{\gamma}^2 = 0.928$ at 290 Hz), the passive pair stays near zero. (b) Per-frame RMS: CH1 210 mV (driven), CH2 54 mV, CH3 0.45 mV. The 52.6 dB gap shows the pipeline cleanly separates a real signal from a silent channel.

8.3 Path Towards Detectability

In order of expected impact: (1) low-noise pre-amplifier (≈ 18 dB); (2) hardware 1 kHz high-pass filter to kill $1/f$ and mains (≈ 20 dB on CH1); (3) resonant 150 kHz piezo with 5 MSa/s ADC to reach the cavitation band; (4) a vibration-isolated pedestal. Steps 1 and 2 together would bring CH1 below 5 mV RMS, within range of strong AE events.

9. Conclusion

We characterised a piezoelectric AE system (calibrated band 5 kHz to 35 kHz, equidistant sensors) and evaluated it on *Euphorbia leuconeura*. A controlled watering experiment yields no detectable AE response after thermal drift correction ($p = 0.86$); the nominally significant Wilcoxon results ($p = 0.002$) are fully explained by thermal drift (Kendall $\tau = +0.69$). A positive control confirms the pipeline detects real signals (52.6 dB, $p < 10^{-12}$), so the negative result is genuine, not a pipeline failure.

The MEMS campaign independently finds no AE evidence: air and soil runs both yield zero median delay, while soil insertion raises the median level to approximately 303 mV and exposes a coherent 32.59 kHz interference comb plus a CH3 saturation fault. Similar channel sensitivity therefore confirms that both MEMS operate, but does not establish biological origin.

The limiting factors remain the LM358 noise floor, common-mode coupling, insufficient upper bandwidth, and incomplete geometry metadata. A low-noise pre-amplifier plus a hardware high-pass filter should lower the piezo floor by ≈ 38 dB; future MEMS runs additionally require recorded sensor distances, galvanic/common-mode controls, and a time-locked positive impulse before passive candidates can be attributed to the plant.

Data and code. All measurement data, analysis notebooks, and this paper's figure generation script are in the project repository at `src/characterization/`. Historical figures are reproducible via `python3 paper/generate_figures.py`; the MEMS frequency calibration and noise floor (Fig. 4) is reproduced from the raw git-LFS recordings via `python3 scripts/analyze_mems_phase3.py`, and further results and plots reside with runs.

References

Y. Benjamini and Y. Hochberg (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J. R. Stat. Soc. B*, 57(1):289–300.

L. Bonisoli, L. Forti, and L. Arru (2025). Outdoor detection of plant ultrasonic emissions using a contactless microphone. *Sensors*.

H. Cochard, P. Cruiziat, and M. T. Tyree (1992). Use of positive pressures to establish vulnerability curves. *Plant Physiol.*, 100(1):205–209.

L. De Roo, L. L. Vergeynst, N. J. F. De Baerdemaeker, and K. Steppe (2016). Acoustic emissions to measure drought-induced cavitation in plants. *Appl. Sci.*, 6(3):71.

M. Gagliano, S. Mancuso, and D. Robert (2012). Towards understanding plant bioacoustics. *Trends Plant Sci.*, 17(6):323–325.

I. Khait et al. (2023). Sounds emitted by plants under stress are airborne and informative. *Cell*, 186(7):1328–1336.

A. Király, D. Farkas, and J. Dobránszki (2025). Ultrasound in plant life and its application perspectives in horticulture and agriculture. *Plants*.

J. Klaminder, M. M. Rollo Jr, T. Näsholm, and J. Viklund (2025). Ultrasonic acoustic emissions as indicators of tree drought stress in outdoor forest settings. *For. Ecol. Manag.*

C. H. Knapp and G. C. Carter (1976). The generalized correlation method for estimation of time delay. *IEEE Trans. Acoust. Speech Signal Process.*, 24(4):320–327.

J. A. Milburn and R. P. C. Johnson (1966). The conduction of sap. II. Detection of vibrations produced by sap cavitation in *Ricinus* xylem. *Planta*, 69(1):43–52.

A. Nardini, M. Battistuzzo, and T. Savi (2013). Shoot desiccation and hydraulic failure in temperate woody angiosperms during an extreme summer drought. *New Phytol.*, 200(1):322–329.

D. Oletić, S. Rosner, M. Zovko, and V. Bilas (2020). Time-frequency features of grapevine's xylem acoustic emissions for detection of drought stress. *Comput. Electron. Agric.*, 178:105787.

D. Oletić, S. Rosner, and V. Bilas (2023). Field-experiences of tracking plant's xylem embolism formation with embedded acoustic emission sensors. *e-J. Nondestruct. Test.*, 28(3).

B. Saha and A. Rastogi (2026). Plant acoustic emission as early stress signals: towards remote integrated monitoring for sustainable agriculture. *Eur. J. Agron.*

D. S. Simbeye, M. E. Mkiramweni, and B. Karaman (2023). Plant water stress monitoring and control system. *Smart Agric. Technol.*, 5:100196.

M. T. Tyree and M. A. Dixon (1983). Cavitation events in *Thuja occidentalis* L.? Ultrasonic acoustic emissions from the sapwood can be measured. *Plant Physiol.*, 72(4):1094–1099.

R. Zweifel and R. Häsler (2001). Dynamics of water storage in mature subalpine *Picea abies*. *Tree Physiol.*, 21(9):561–569.